

Note 01: Runtime

Instructions

We are going to determine whether one of the three standard sorting algorithms is fastest, or whether they all perform at practically the same speed.

Background

The three standard sorting algorithms are bubble sort, insertion sort, and selection sort. Bubble sort repeatedly swaps adjacent elements that are out of order until the entire collection is sorted. Insertion sort builds its sorted portion of the collection one item at a time, inserting each new item into its correct position within its sorted portion. Selection sort repeatedly finds the minimum (or maximum, for descending order) value in its unsorted portion of the collection and swaps it into the first position of its unsorted portion.

Procedures

1. Define each sorting algorithm. As a comment, calculate each of their runtime functions.
2. In the main function, create three copies of a randomly generated array with 5000 elements.
3. Calculate how long each algorithm takes to sort the array.